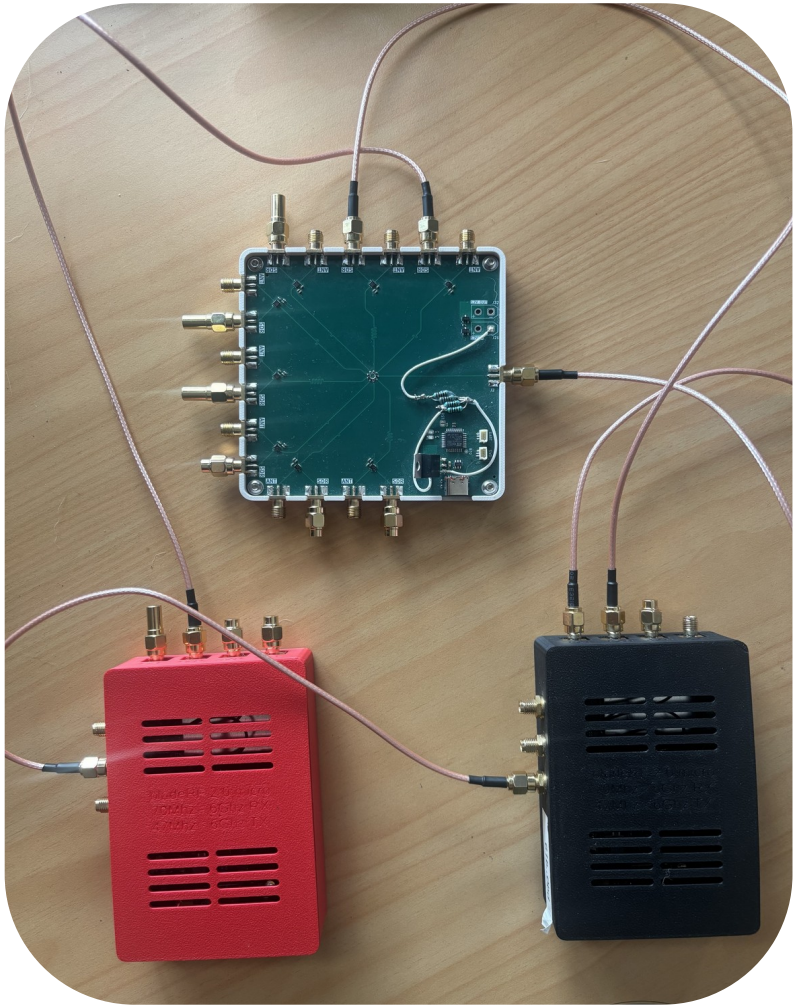
Angle of Arrival Estimation using Two SDRs



Motivation

Rocket localisation: Develop direction-finding capabilities to support rocket tracking.

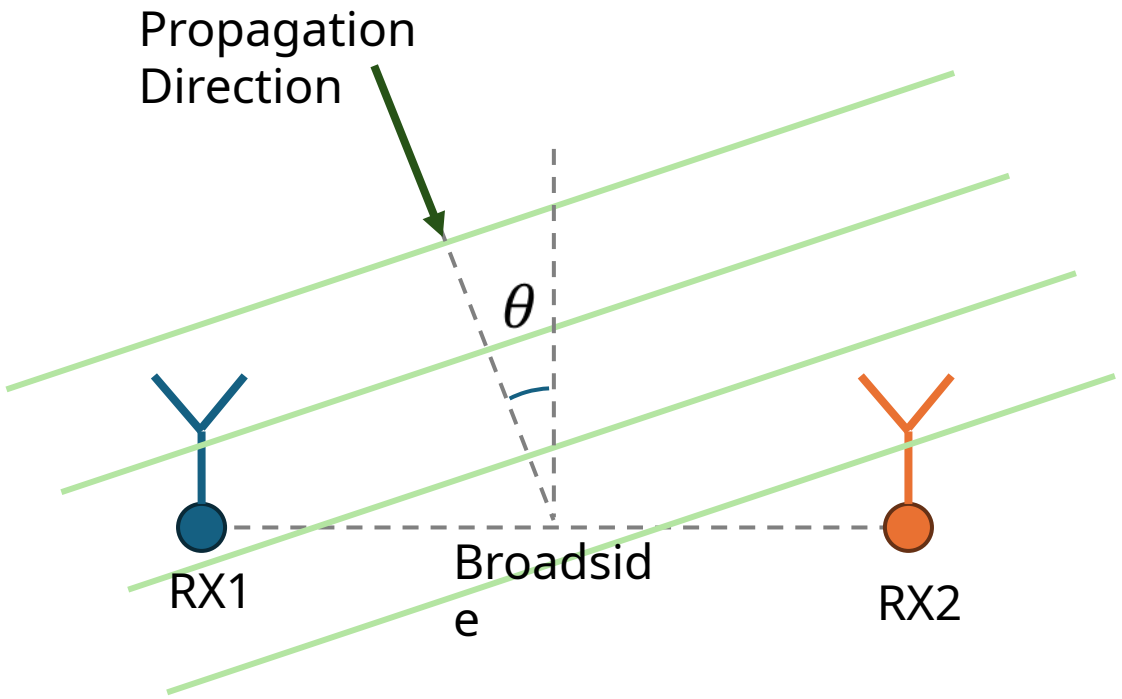
UC Rocketry: Make use of existing bladeRF hardware to create a system that can be modified and expanded.

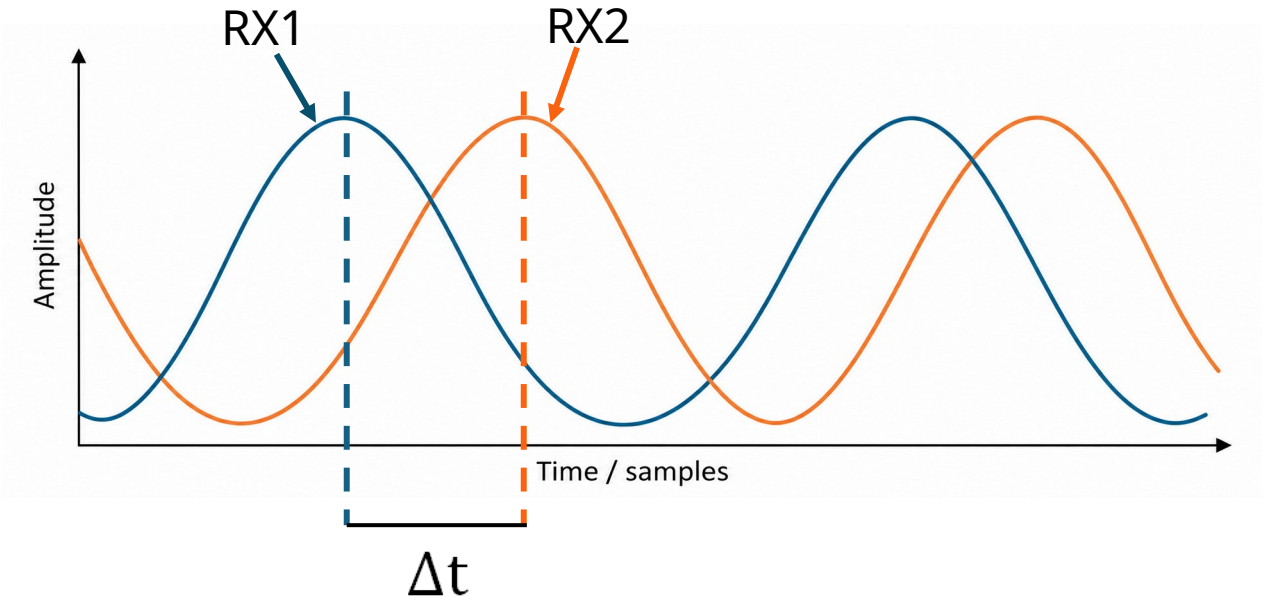
Project Aim

Develop and validate a dual-SDR system capable of synchronised, calibrated phase measurements for AoA estimation.

Chanelle Duffy

Angle of Arrival Estimation Using Two SDRs





Incoming wavefront reaches RX1 first

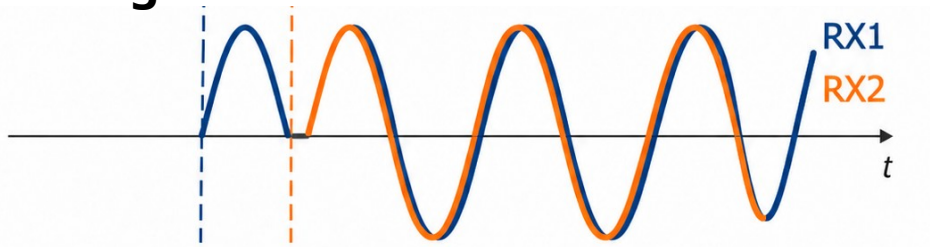
Received Signals are Phase Shifted

The Challenge of Independent SDRs

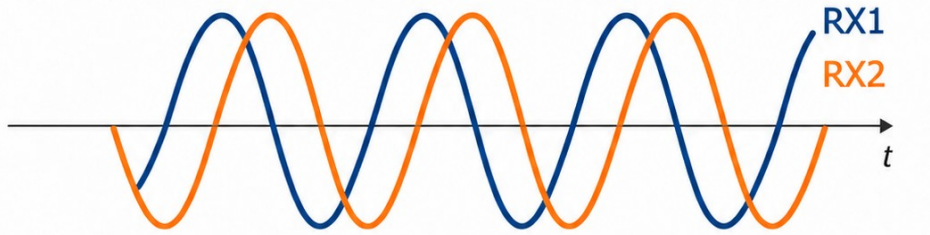
Offset	Primary Cause
Timing	Independent sample acquisition start times
Phase	RF signal paths and independent PLL
Amplitude	Receiver gain and component differences

$$\Delta\phi_{measured} = \Delta\phi_{AoA} + \Delta\phi_{system}$$

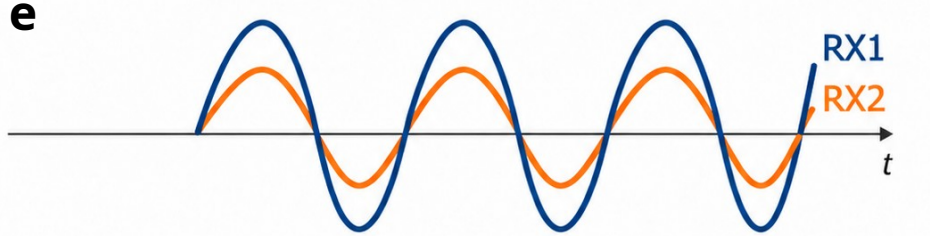
Timing



Phase



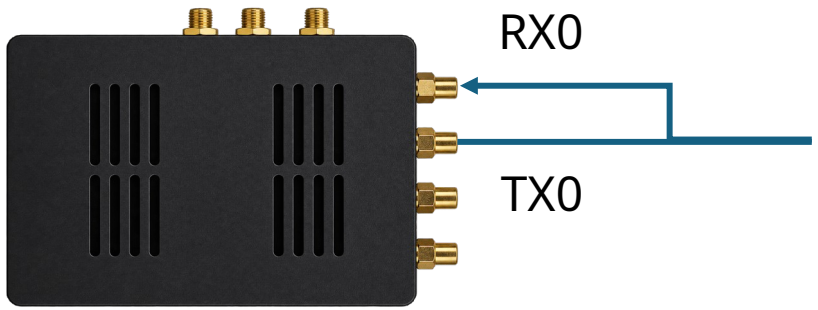
Amplitud e



Initial Offset Calibration

Calibration → Dual SDR Synchronisation → RF Switch Board → Angle-of-Arrival (AoA)

Hardware Configuration



Master SDR

Purpose

- Establish a method for estimating and correcting relative timing, amplitude, and phase offsets.

Section Overview

- Calibration algorithm development
- Single-SDR hardware testing

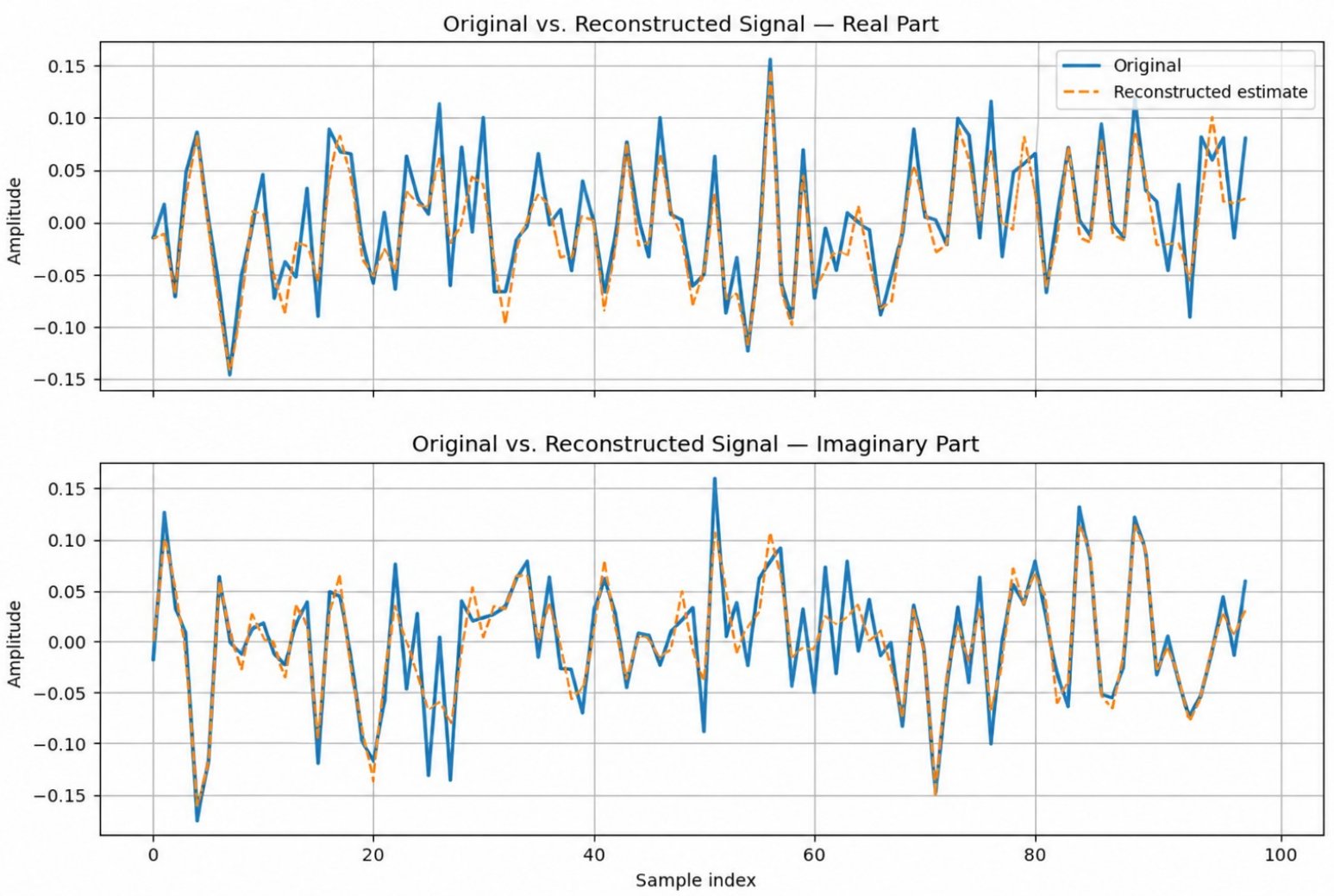
Initial Calibration Development

Cross-Correlation
Estimate the integer sample delay

Phase-Fitting
Estimate the fractional delay

Least-Squares Complex Gain
Estimate relative amplitude and phase

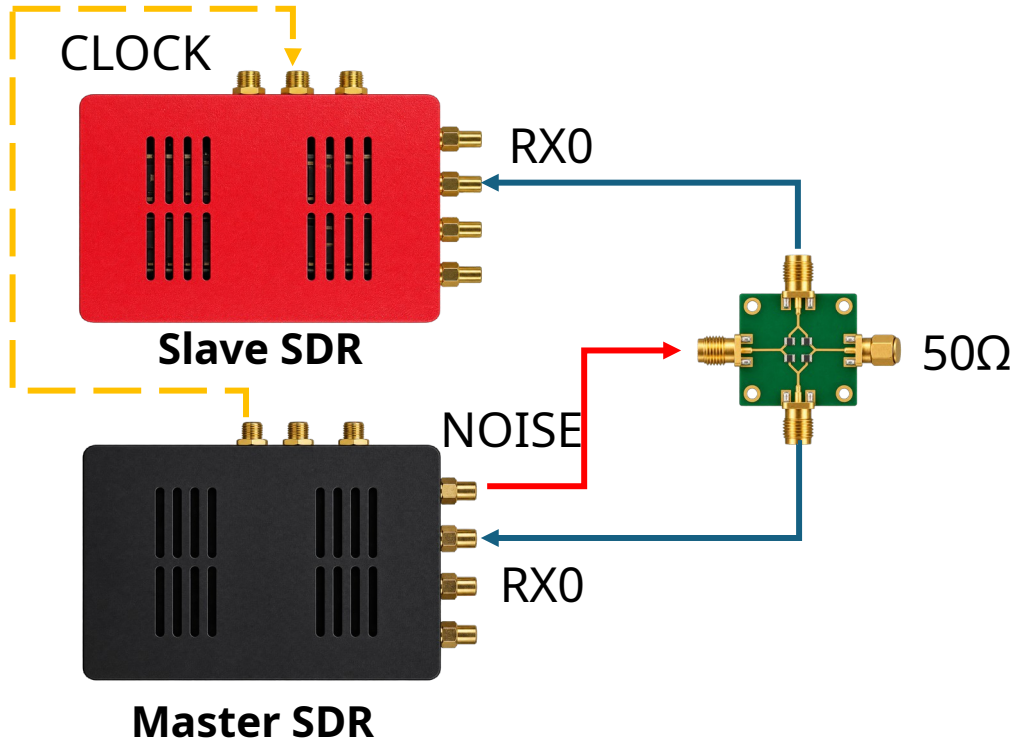
Apply Correction
Align received IQ signal to the reference



SDR Synchronisation

Calibration → **Dual SDR Synchronisation** → RF Switch Board → Angle-of-Arrival (AoA)

Hardware Configuration



Purpose

- Extend calibration to two independent SDR receivers.
- Maintain relative alignment during continuous acquisition.

Section Overview

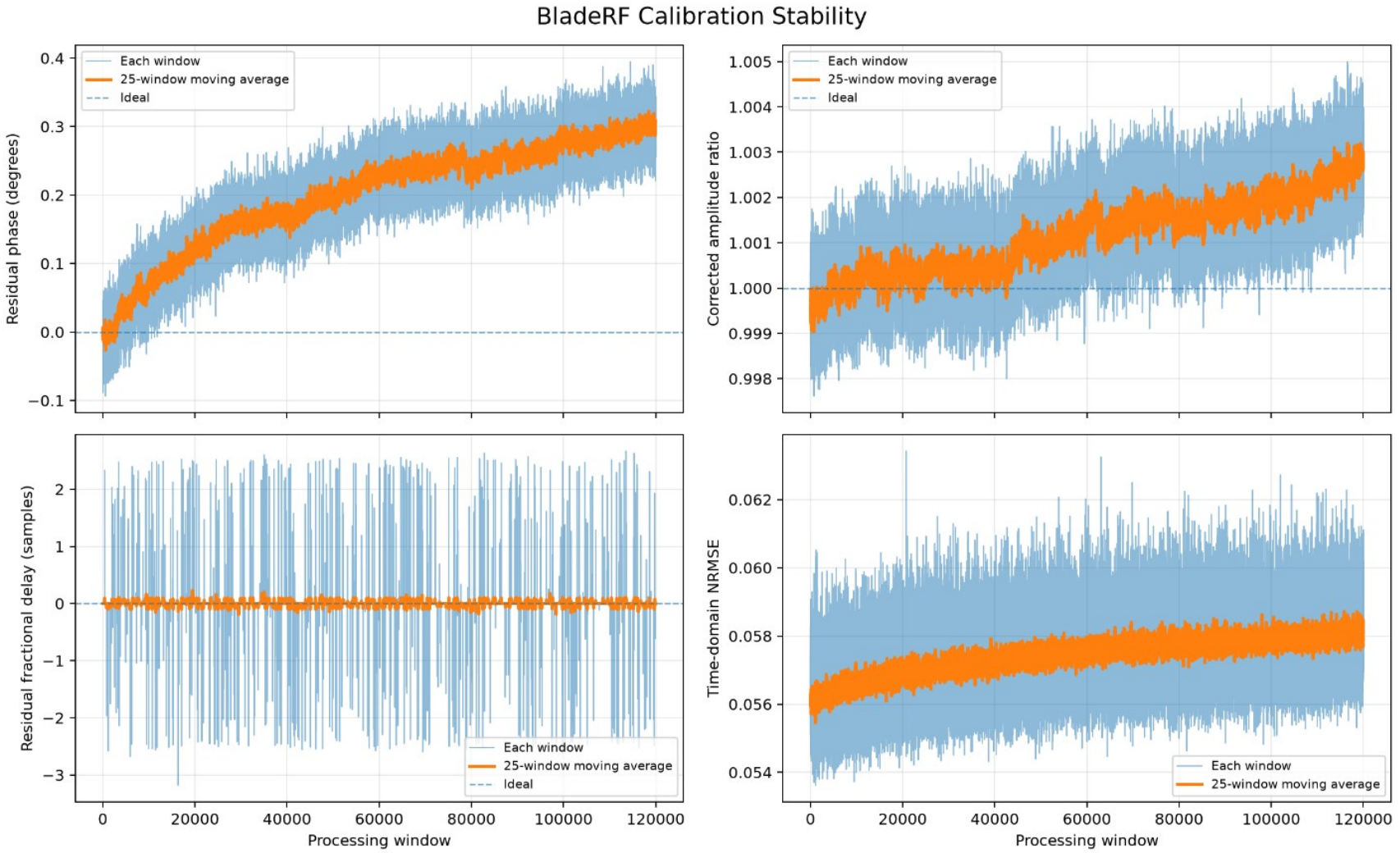
- Shared Clock Synchronisation
- Dual-SDR acquisition and calibration
- Continuous correction and stability testing.

SDR Synchronisation

1. Shared Reference Clock
Master CLKOUT → Slave CLKIN
Common 38.4 MHz frequency ref

2. Initial Sample Alignment
Apply estimated integer delay

3. Continuous Correction
Fractional-delay FIR + complex gain



Improving Calibration Reliability

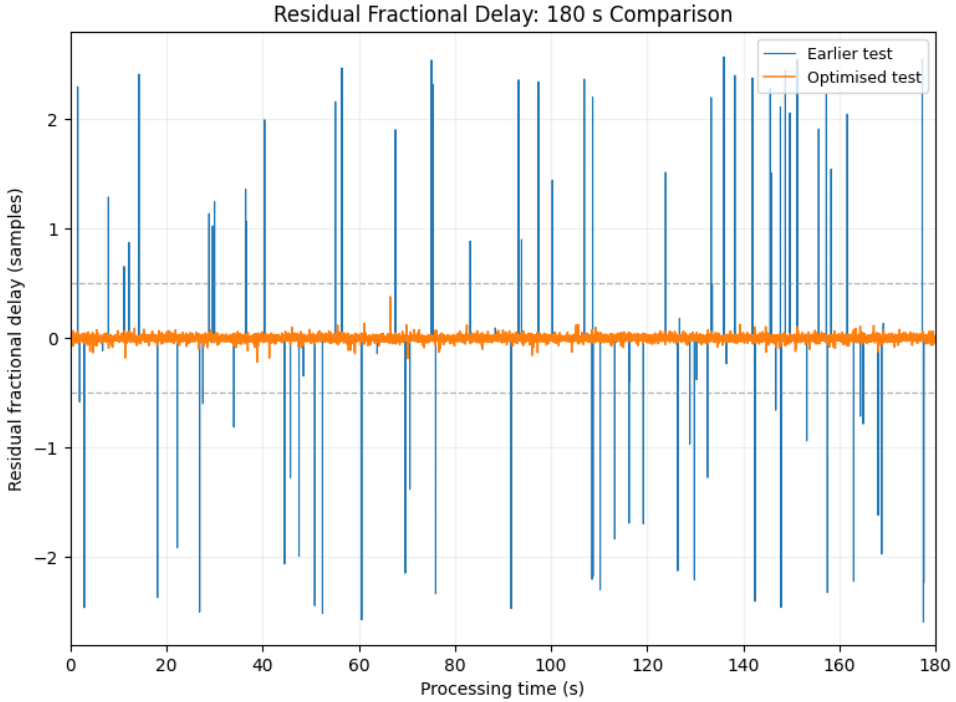
1. Fractional Delay Estimation

Original phase-unwrapping method

0.1307 Sample Standard Deviation

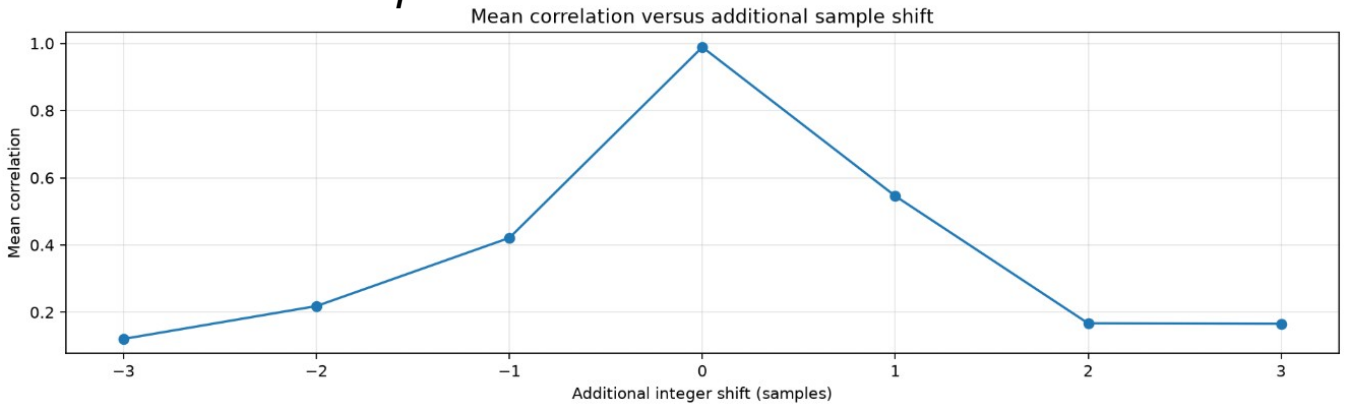
Bounded Correlation Optimisation

0.0170 Sample Standard



2. Integer Delay Reliability

Original Peak Estimate +/- 2 Samples



3. Dynamic Frequency Masking

Retains bins with ≥ 6dB above the estimated noise floor.

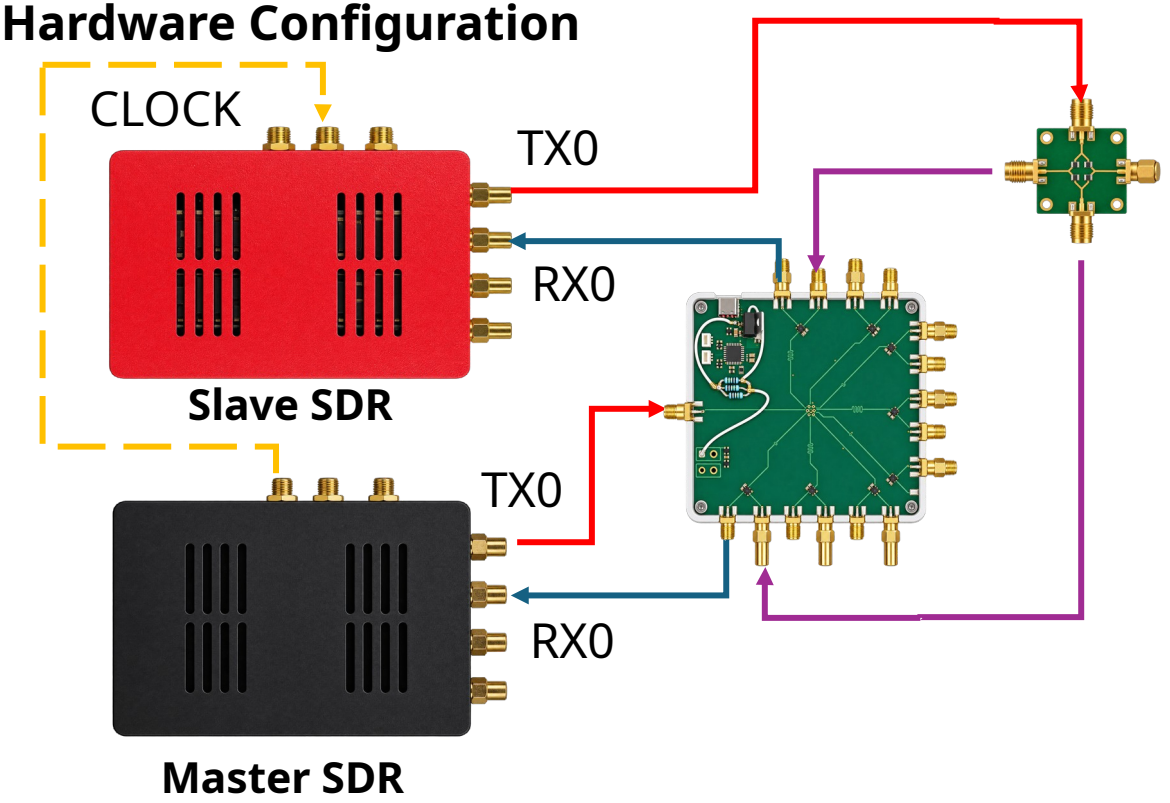
±250/300/350 kHz Bandwidths validated

0.9906	18,000/18,000	180 s
Mean Correlation	Zero integer delays	Trial time

RF Switch Board Integration

Calibration → Dual SDR Synchronisation → **RF Switch Board** → Angle-of-Arrival (AoA)

Hardware Configuration



Purpose

- Validate RF signal paths
- Assess whether calibration holds after switching between RF paths.

Section Overview

- Individual RF path validation
- Dual-SDR integration and calibration
- Switching repeatability and phase stability

RF Switch Board Integration & Validation

1. Individual Path Validation
Verified NOISE and ANTENNA paths separately

↓

2. Dual SDR Integration
Verified calibration through the board

↓

3. Switching Repeatability
NOISE → ANTENNA → NOISE

Switching Test Results

	10 Switches	No Switch (180s)
Correlation	0.99893	0.99901
Phase Change	+0.310°	+0.202°

Run-to-Run Phase Repeatability

10 independent starts, NOISE → ANTENNA

15.084°

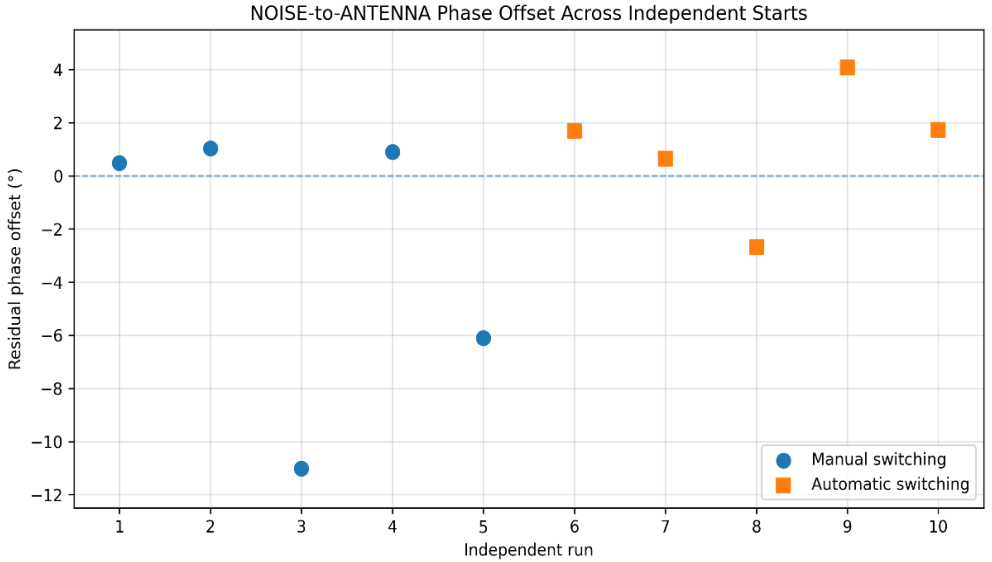
Total residual phase offset spread

-11.02 ⟷ +4.07

Mean correlation: 0.9989

Residual Phase Offset Investigation

Experiment	Result	Finding
1. Independent RF Source Replaced slave TX with an external noise generator.	**8.635°** Phase offset spread across 5 starts	Variation persisted without the slave transmitter
2. Master TX Shutdown Disabled master TX while in ANTENNA mode.	**+0.2° to +0.3°** Difference in phase after stopping TX	A consistent small change – not enough to account for observed offsets
3. Single SDR Dual-Channel Connected both RF switch outputs to RX0 and RX1 of a single bladeRF.	**0.177°** Difference	The issue arises within the dual SDR setup.



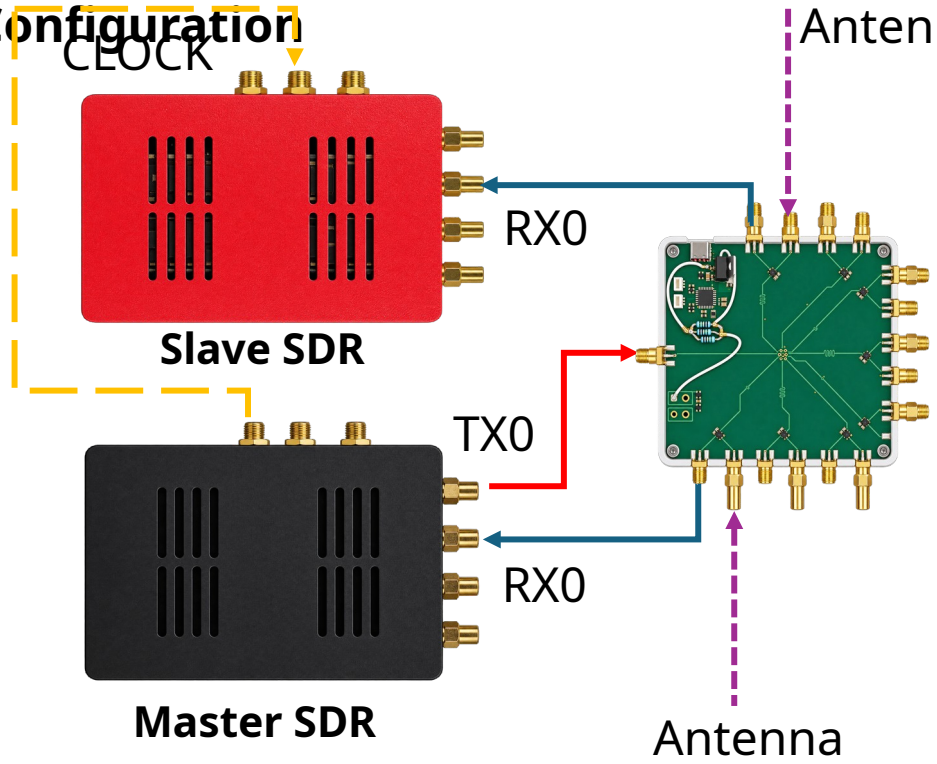
Conclusion:

The underlying cause of the run-to-run phase offset was not determined.

Angle of Arrival

Calibration → Dual SDR Synchronisation → RF Switch Board → **Angle-of-Arrival (AoA)**

Recommended Hardware Configuration



Purpose

- Develop AoA estimation using relative phase
- Validate AoA estimation

Section Overview

- Phase difference AoA algorithm development
- Simulation and Monte Carlo
- Residual phase-offset effect

Angle-of-Arrival Estimation and Validation

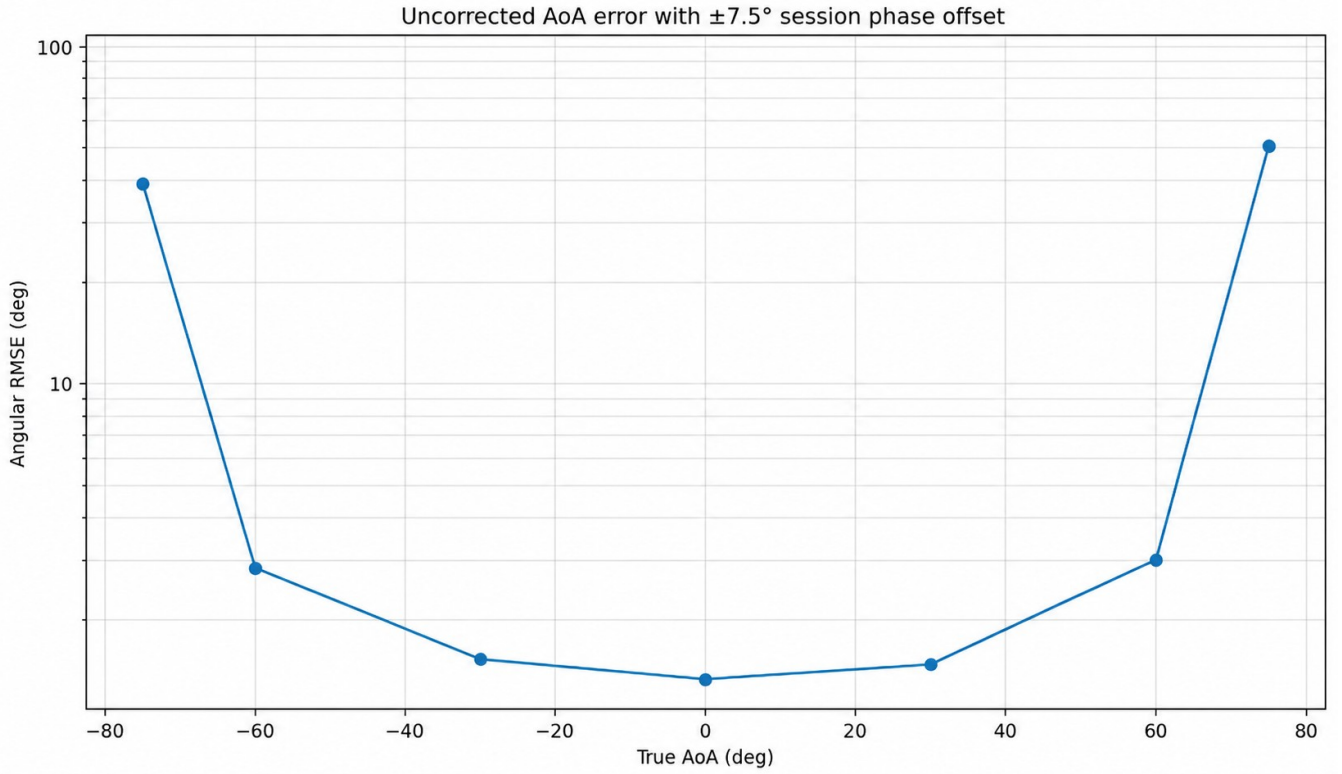
AoA Estimation Method

1. Estimate relative phase

$$\Delta\phi = \arg\left(\sum_{n=1}^{N} x_n^* y_n\right)$$

2. Convert phase to arrival angle

$$\theta = \sin^{-1}\left(-\frac{\lambda\Delta\phi}{2\pi d}\right)$$



Sim Results in RMSE

Angle = 30°

SNR =

0.070°

Ideal conditions: **No calibration error**

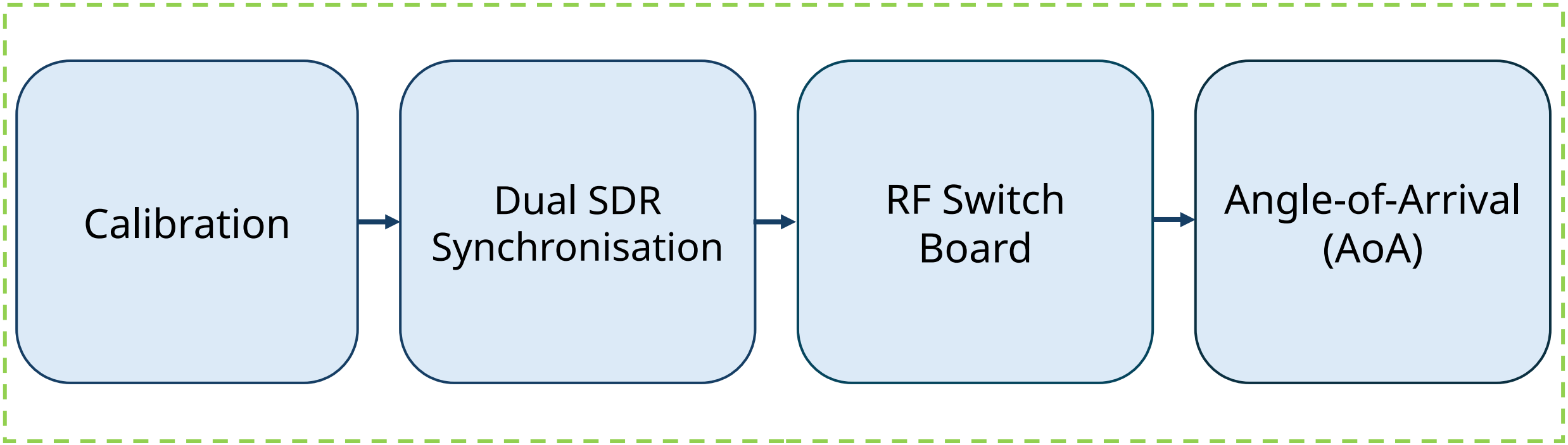
1.477°

Uniform ±7.5° phase offset

0.145°

Uniform ±7.5° phase offset – Post calibrated

Full Pipeline Integration



Complete Pipeline Integration and Validation

1. NOISE-Mode Calibration
Estimate and validate receiver offsets

↓

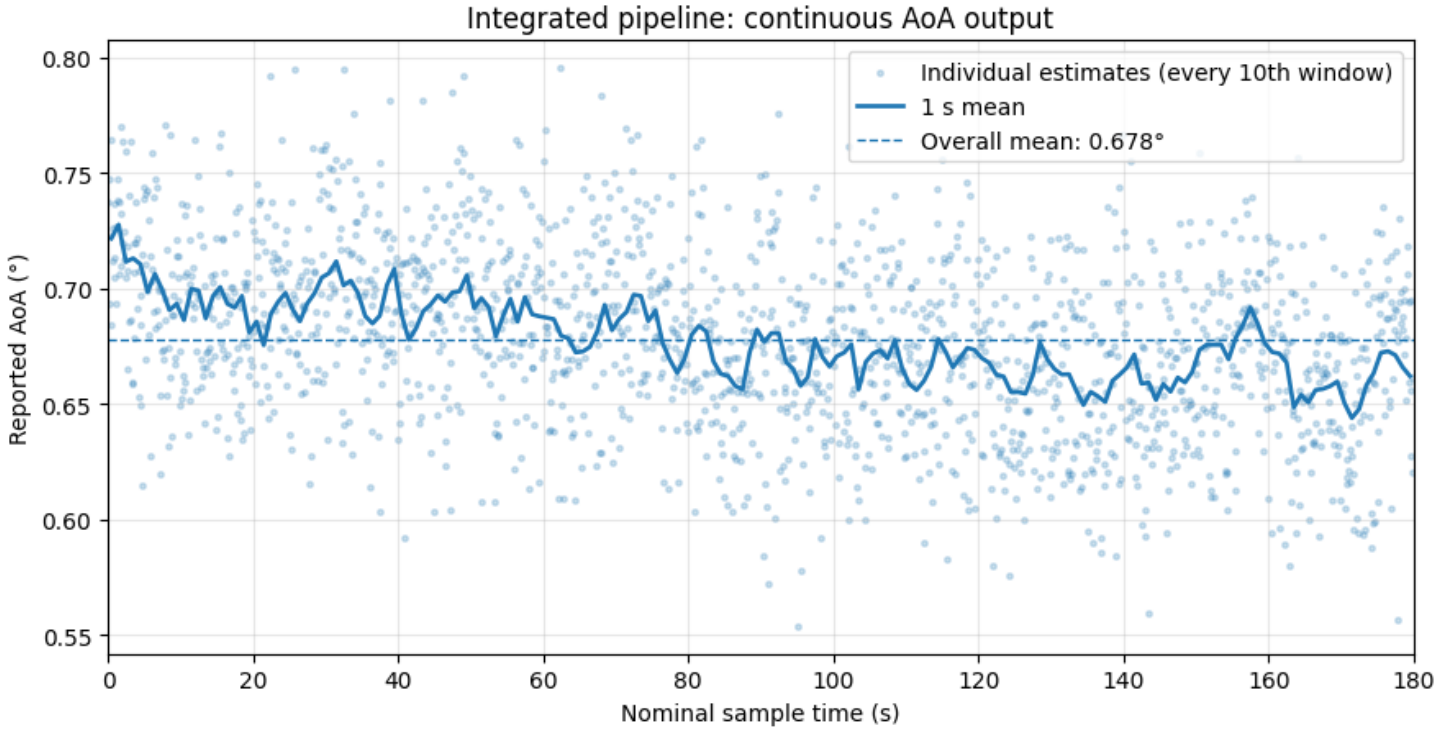
2. RF Path Switching
Disable NOISE TX, switch to ANTENNA

↓

3. Continuous IQ Correction
Apply saved calibration

↓

4. AoA Estimation
Calculate phase difference/angle per window



18,000
Valid Windows

0.9740
Mean Correlation

0.6775°
Mean AoA*

0.0381°
AoA Standard Dev

Conclusion and Project Outcomes

1. Calibration

Developed and validated timing, amplitude and phase correction.

2. Synchronisation

Achieved continuous, calibrated data reception.

3. RF Switching

Integrated and validated RF switching between NOISE and ANTENNA.

4. Angle of Arrival Estimation

Validated in **simulation**, demonstrated complete pipeline.

Remaining Work:

Identify and correct the unknown phase offset, validate AoA accuracy using physical hardware.